



Improving the Fluid Cleanliness through 16M Motor Graders

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Improving the Fluid Cleanliness through 16M Motor Graders.....	1
1.0 Introduction.....	2
2.0 Best Practice Description	3
3.0 Implementation Steps.....	5
4.0 Benefits	5
5.0 Resources Required.....	5
6.0 Supporting Attachments / References	5
7.0 Related Best Practices	5
8.0 Acknowledgements	5

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices.” Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

August 2022
3.01-220810-1

1.0 Introduction

One of the main goals of mining companies around the operations of mobile equipment is to ensure safe conditions, good component lives, and the high productivity of technicians during maintenance.

Most of the manufacturers worked to improve their machine serviceability to avoid component damages and deliver a better bottom line in maintenance. This includes filter relocations, design changes, new models to perform PMs, etc. All of it going to help the people in front of the line about this subject matter.

However, there are innumerable site conditions that have their particularities that affect the technician's mindset to perform maintenance achieving the technical quality levels recommended by Caterpillar.

A Mining Site in Upper North Brazil uses to face high contamination levels into the 16M Tandem, affecting the brakes and oil cleanliness. Also, the shop technician's feedback reported some difficulty to perform breather replace during the preventive maintenance.



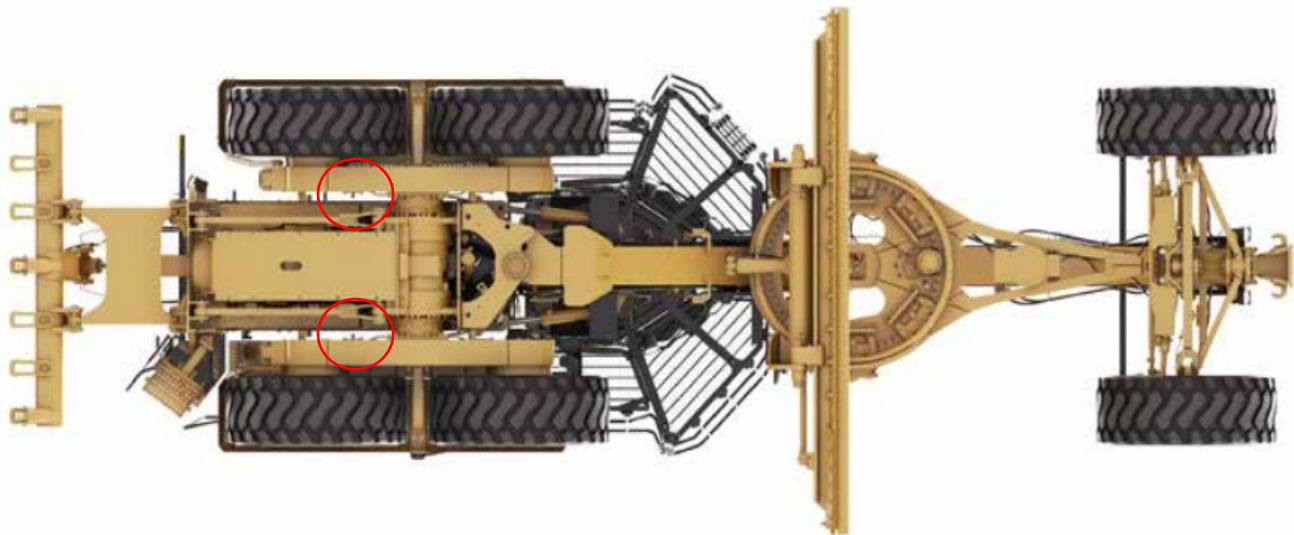
Picture 1 - Breather (4H-6112) place faces contamination around bauxite operations.

This Best Practice:

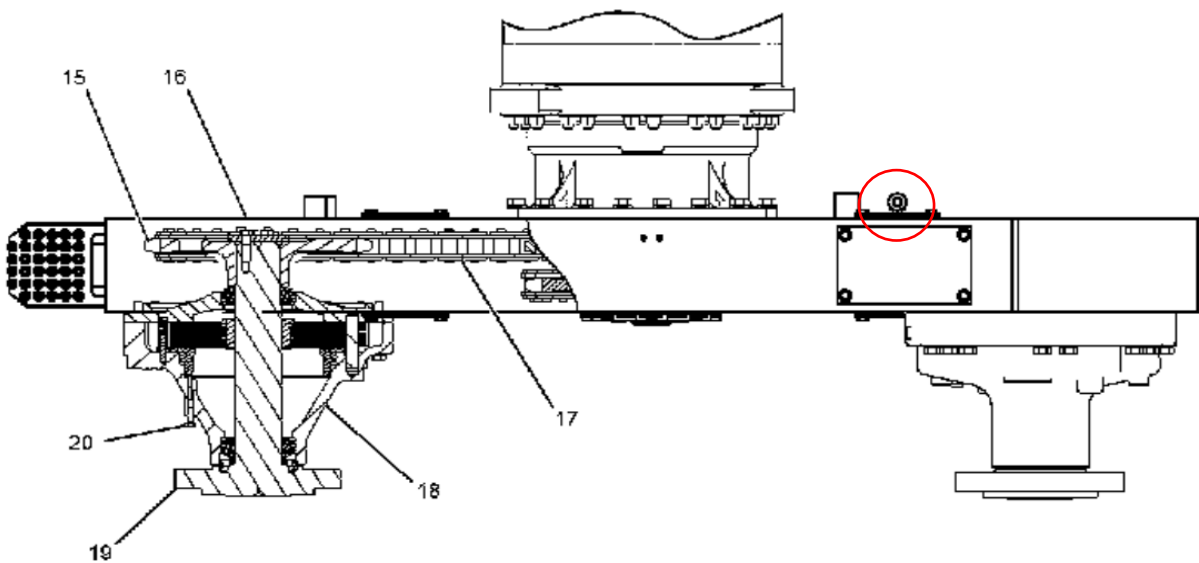
Caterpillar **SEBU9228-17** literature recommends replacing the Breather (4H-6112) on the impossibility of removal for cleaning, or advanced state of damage. It is also recommended a 500-hour period to replace this part during preventive maintenance.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Improving the Fluid Cleanliness Through 16M Motor Graders	DATE 8/9/2022	CHG NO 00	NUMBER 3.01-220810-1

View of the MG-16 Motor Grader:



Breather identification (4H-6112) on the MG 16 Motor Grader under the Machine



Breather (4H-6112) is replaced in Preventive Maintenance every 500 hours of operation.

2.0 Best Practice Description

This Best Practice prevents contamination by various types of debris, promoting changes in the Breather of the Caterpillar MG 16 Motor Grader, which reduces risks to the equipment's integrity, simplifying cleaning and adding dynamism to maintenance.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Improving the Fluid Cleanliness Through 16M Motor Graders	DATE 8/9/2022	CHG NO 00	NUMBER 3.01-220810-1



New Breather position after procedure completion.

These improvements consist of adding an extender made by the SOTREQ Engineering Team and repositioning the Breather (4H-6112) making the material accumulation on the structure more difficult.



Breather new position - CAT MG 16

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Improving the Fluid Cleanliness Through 16M Motor Graders	DATE	CHG NO	NUMBER
	8/9/2022	00	3.01-220810-1

3.0 Implementation Steps

For the procedure of repositioning the breather, Caterpillar literature **PECP5030-11** was used for technical guidance on the execution of the assembly procedures, from a mining customer's need, this Best Practice was developed respecting the existing Caterpillar standardization for compatibility of the fluids and dimensioning of the threads on the terminals.

4.0 Benefits

- Reduced risk of Tandem contamination during routine operation.
- More simple maintenance/replacement processes during preventive maintenance.
- Improved accessibility.
- Decreased time required for maintenance.

5.0 Resources Required

Genuine Caterpillar parts were used to ensure the System's reliability and effectiveness.

Item	Qty (Unity/Length)	PN
Hose	2 m	456-8409
Coupling Perm	01	124-1925
Coupling Permanent	01	124-1891
Clip	02	1S0-994

6.0 Supporting Attachments / References

CAT **PECP5030-11** Literature.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

We would like to thank the Engineering and Product Support team for support in the development of this Best Practice.

We would like to thank Technician Alan S. Martins for their support.

Our special thanks to Doug Friedrich of Caterpillar Inc. For unique technical insights that contribute to the continued evolution of our operations, Thanks Doug.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Improving the Fluid Cleanliness Through 16M Motor Graders	DATE 8/9/2022	CHG NO 00	NUMBER 3.01-220810-1

Bruno Cuimar

Sotreq S/A

Fleet Performance Engineer

E-mail: bruno.cuimar@sotreq.com.br

Mobile: +55 091 3211-9610 / +55 91 9265-6029

Doug Friedrich

Caterpillar Inc.

Site Performance Manager

E-mail: friedrich_douglas_w@cat.com

Mobile: +1.305.807.9542

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Improving the Fluid Cleanliness Through 16M Motor
Graders

DATE
8/9/2022

CHG
NO
00

NUMBER
3.01-220810-1